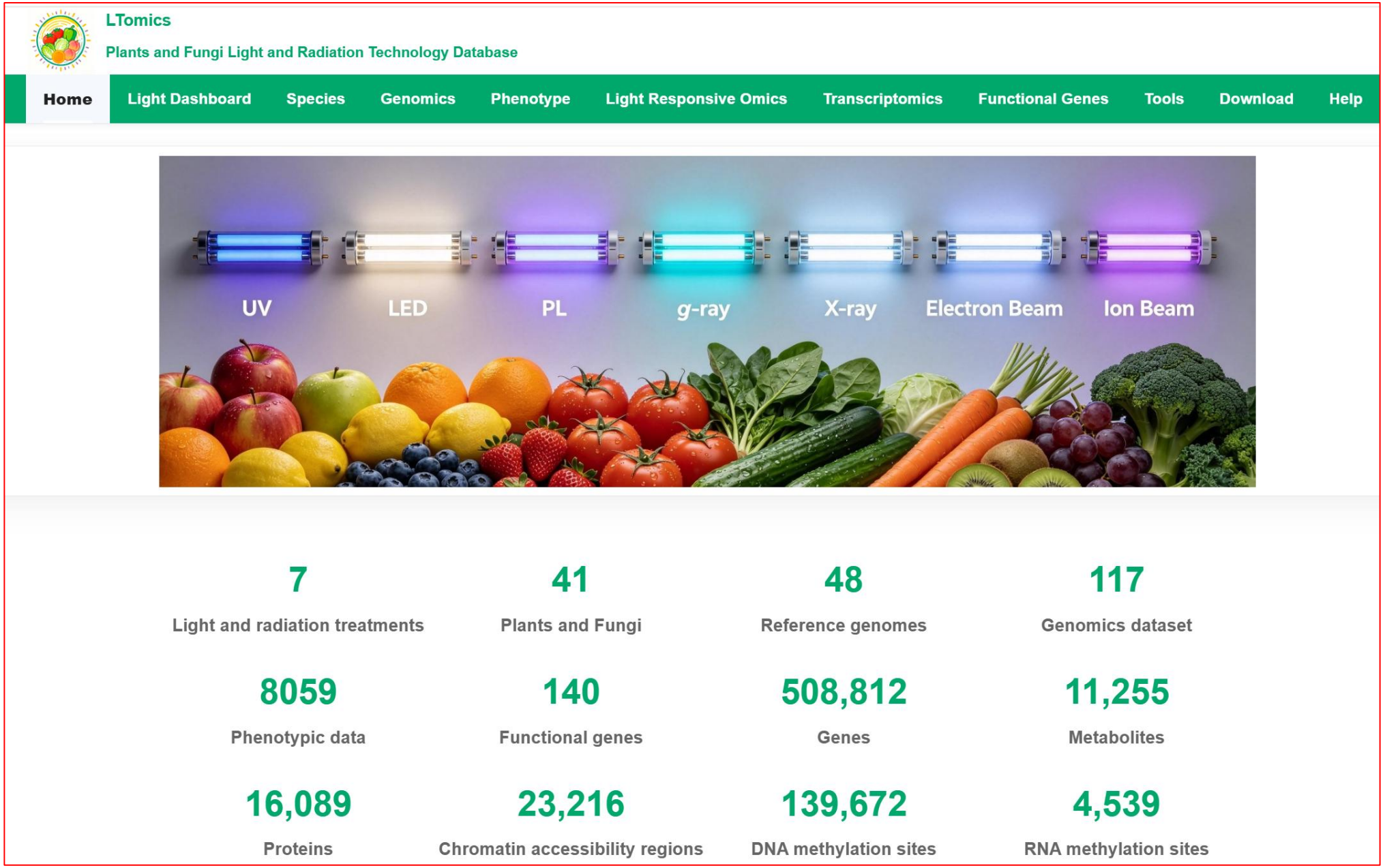
LTomics User Manual

URL：https://fvfreshlighttech.biocloud.net/

1. Home page


LTomics
Plants and Fungi Light and Radiation Technology Database
Home
Light Dashboard
Species
Genomics
Phenotype
Light Responsive Omics
Transcriptomics
Functional Genes
Tools
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LED
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Reference genomes
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8059
Phenotypic data
140
Functional genes
508,812
Genes
11,255
Metabolites
16,089
Proteins
23,216
Chromatin accessibility regions
139,672
DNA methylation sites
4,539
RNA methylation sites

2. Light Dashboard module

Light Dashboard

View detailed help



UV

Ultraviolet (UV) radiation (10–400 nm) is divided into UV-A (315–400 nm), UV-B (280–315 nm), and UV-C (100–280 nm), with UV-B and UV-C exhibiting the most significant biological effects on plants. Pre-harvest UV treatment imposes mild stress on growing plants, establishing a "stress memory" that pre-activates defense and metabolic pathways, thereby increasing phenolic and flavonoid contents, promoting early flowering, and enhancing disease resistance. Post-harvest UV treatment acts directly on harvested produce by activating antioxidant enzymes, inhibiting microbial growth, and retarding chlorophyll degradation and fruit softening, effectively reducing decay and weight loss while preserving vitamin C and soluble solids, thus extending shelf life. Complementary pre- and post-harvest UV applications offer synergistic potential to maximize the commercial value of horticultural products.

At the top of Light Dashboard module, a detailed introduction to the seven types of light and ionizing radiation treatments is provided.

2. Light Dashboard module

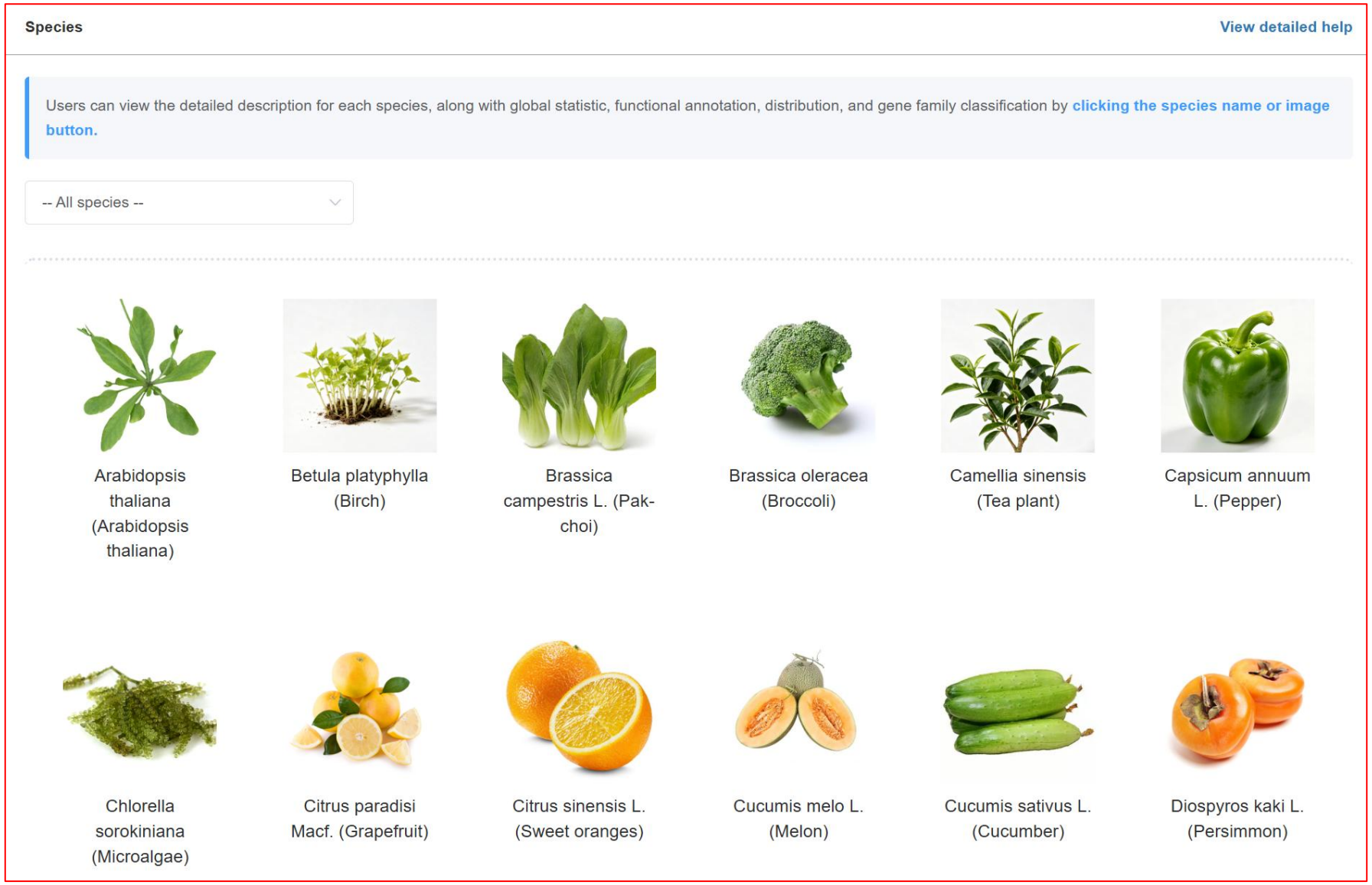
Download

Number	Name	Species	Cultivar	Pre/Post	Light type	Treatment dosage and administration	Omics
1	Grape	Vitis vinifera	Zaotian	Pre	LED	Artificial light treatment (LED red and blu…	ND1, D
2	Grape	Vitis vinifera	Kyoho	Pre	LED	200 ± 5 μmol·m-2·s-1 white, blue, red, g…	W15, V
3	Grape	Vitis vinifera	Pinot Noir	Post	UV	0.9 mJ·cm-2 UV-C	Dark, U
4	Grape	Vitis vinifera	Cabernet Sauvigno	Pre	UV	0.1 W·m-2 and 0.3 W·m-2 UV-B treatment	HFG-U
5	Grape	Vitis vinifera	Tonghua-3	Post	UV	300 mW·cm-2 UV-C treatment	CK-4h,
6	Grape	Vitis vinifera	Jumeigui	Post	UV	0.15 W·m-2 UV-C treatment	pos-yg
7	Grape	Vitis vinifera		Post	UV	11 nm UV-B for 15 min	Control
8	Grape	Vitis vinifera	Hongbaladuo	Post	UV	10 min to 6 W·m-2 UV-C	Control
9	Peach	Prunus persica L.	Xiahui 5	Post	UV	4 kJ·m-2 UV-C treatment	Control
10	Peach	Prunus persica L.	Lumi1	Pre	UV	Natural light group, low-dose UVB radiat…	CK, UV

共 80 条 10条/页 1 2 3 4 5 6 ··· 8 前往 1 页

Below Light Dashboard module, detailed information on the 80 treatment studies is provided, which can be sorted and filtered by column names such as "Species", "Light type", and "Preharvest/Postharvest".

3. Species module



Species module displays 41 plant and fungal species.

3. Species module

Basic Information



Solanum lycopersicum (Tomato)

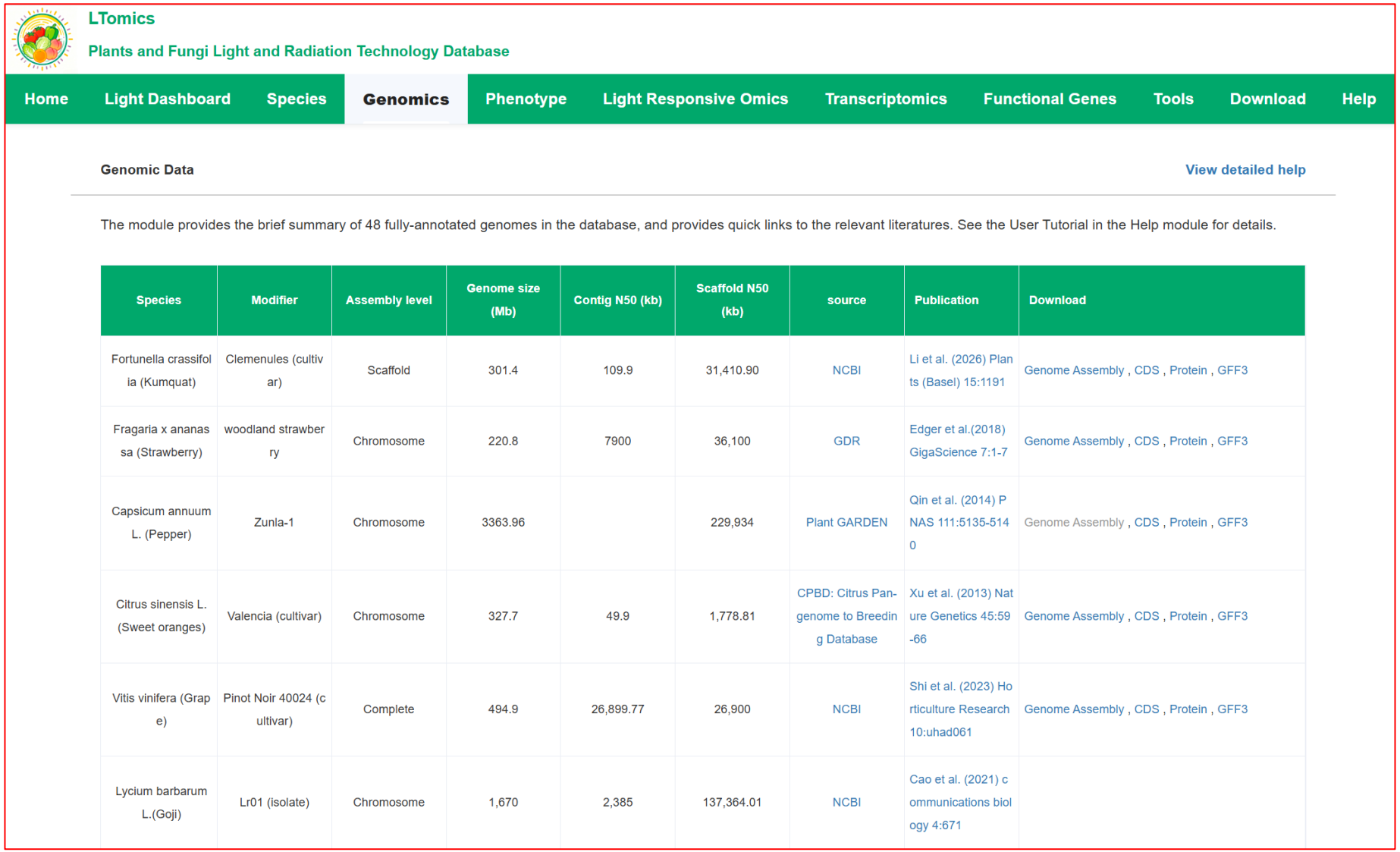
Solanum lycopersicum (Tomato) is an annual or perennial herb of the genus Solanum in the family Solanaceae. Its fruit is typically globose to slightly flattened with smooth, red, yellow, or orange skin and juicy, sweet-tart flesh containing many small seeds. It originates in western South America (from Colombia to Chile).

Clicking on any species image allows you to view its Basic Information and Omics Data Portal. Clicking on any omics name within the Omics Data Portal will direct you to the corresponding omics page.

Omics Data Portal

Genome | mRNA | Protemics | ATAC | RNA Methylation | RNA Acetylation

4. Genomics module—Genomic Data



LTomics

Plants and Fungi Light and Radiation Technology Database

Home | Light Dashboard | Species | Genomics | Phenotype | Light Responsive Omics | Transcriptomics | Functional Genes | Tools | Download | Help

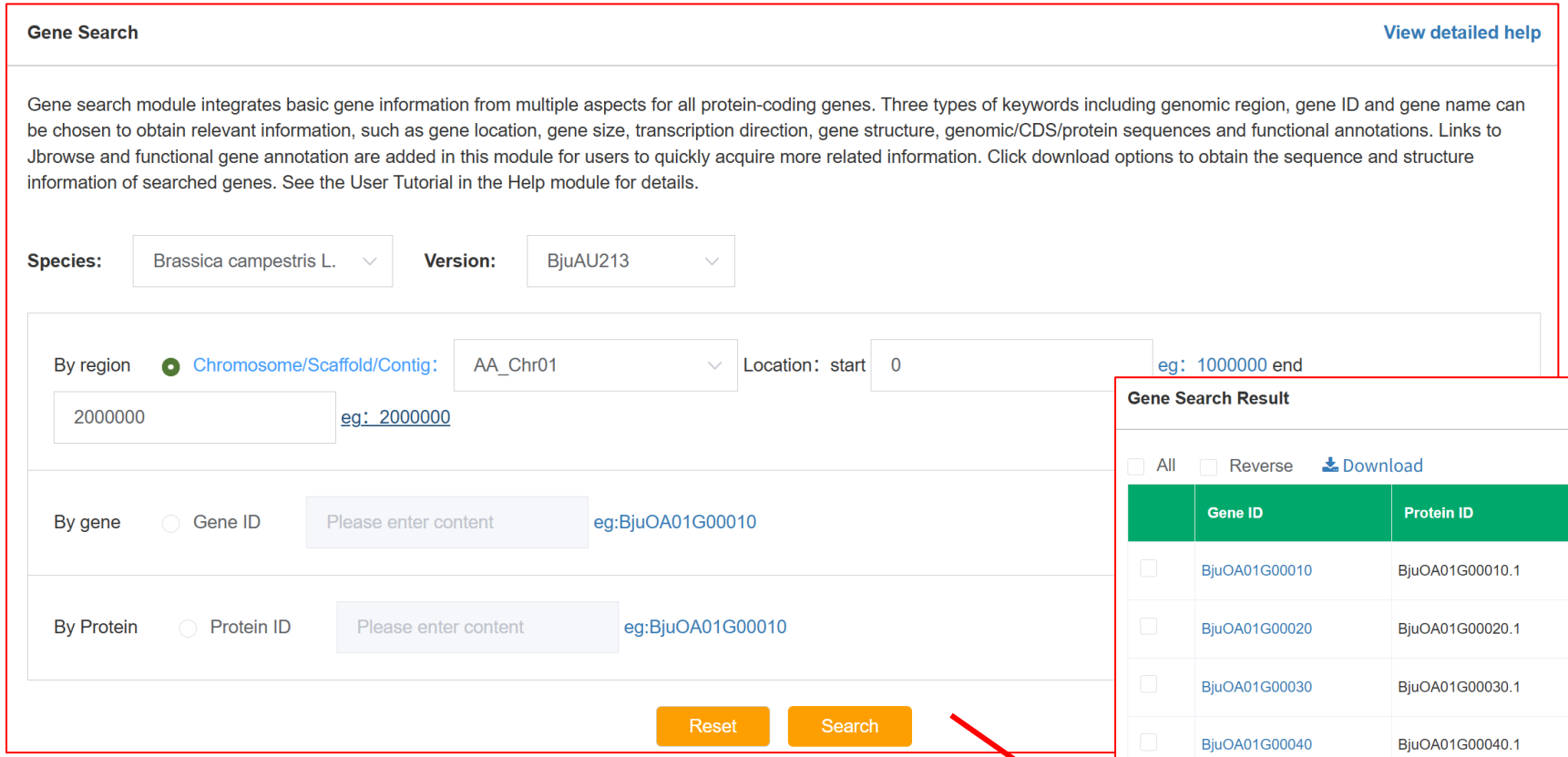
Genomic Data View detailed help

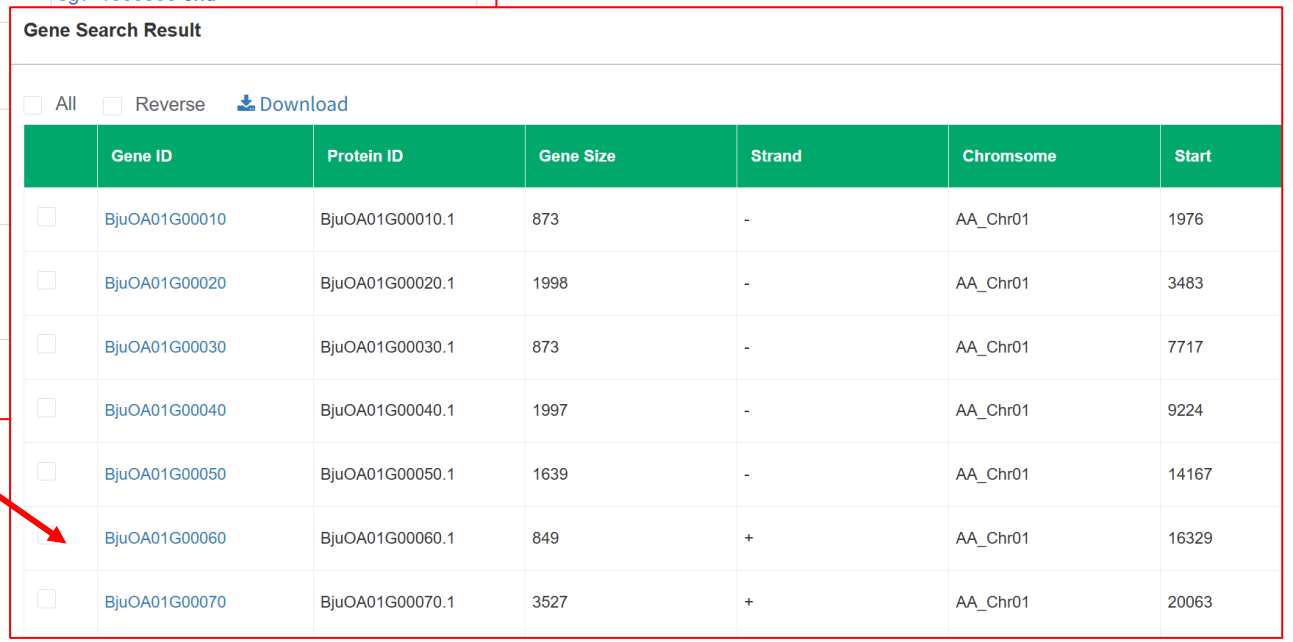
The module provides the brief summary of 48 fully-annotated genomes in the database, and provides quick links to the relevant literatures. See the User Tutorial in the Help module for details.

Species	Modifier	Assembly level	Genome size (Mb)	Contig N50 (kb)	Scaffold N50 (kb)	source	Publication	Download
Fortunella crassifolia (Kumquat)	Clemenules (cultivar)	Scaffold	301.4	109.9	31,410.90	NCBI	Li et al. (2026) Plants (Basel) 15:1191	Genome Assembly , CDS , Protein , GFF3
Fragaria x ananassa (Strawberry)	woodland strawberry	Chromosome	220.8	7900	36,100	GDR	Edger et al.(2018) GigaScience 7:1-7	Genome Assembly , CDS , Protein , GFF3
Capsicum annuum L. (Pepper)	Zunla-1	Chromosome	3363.96		229,934	Plant GARDEN	Qin et al. (2014) PNAS 111:5135-5140	Genome Assembly , CDS , Protein , GFF3
Citrus sinensis L. (Sweet oranges)	Valencia (cultivar)	Chromosome	327.7	49.9	1,778.81	CPBD: Citrus Pan-genome to Breeding Database	Xu et al. (2013) Nature Genetics 45:59-66	Genome Assembly , CDS , Protein , GFF3
Vitis vinifera (Grape)	Pinot Noir 40024 (cultivar)	Complete	494.9	26,899.77	26,900	NCBI	Shi et al. (2023) Horticulture Research 10:uhad061	Genome Assembly , CDS , Protein , GFF3
Lycium barbarum L.(Goji)	Lr01 (isolate)	Chromosome	1,670	2,385	137,364.01	NCBI	Cao et al. (2021) communications biology 4:671	

The Genomic Data section within the Genomics module provides detailed information for 48 genomes, with links to the corresponding download pages and published literature.

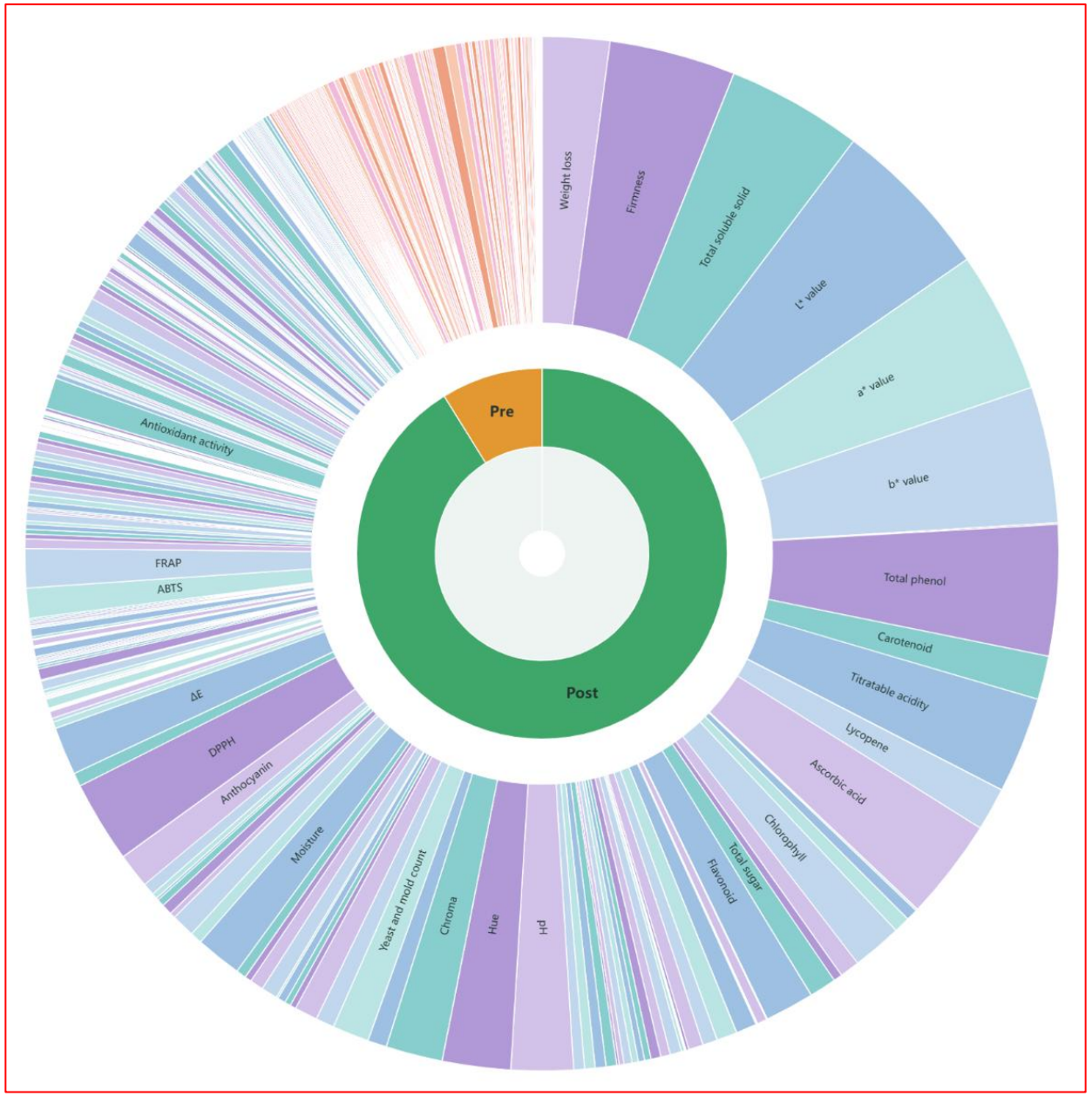
4. Genomics module—Gene Search





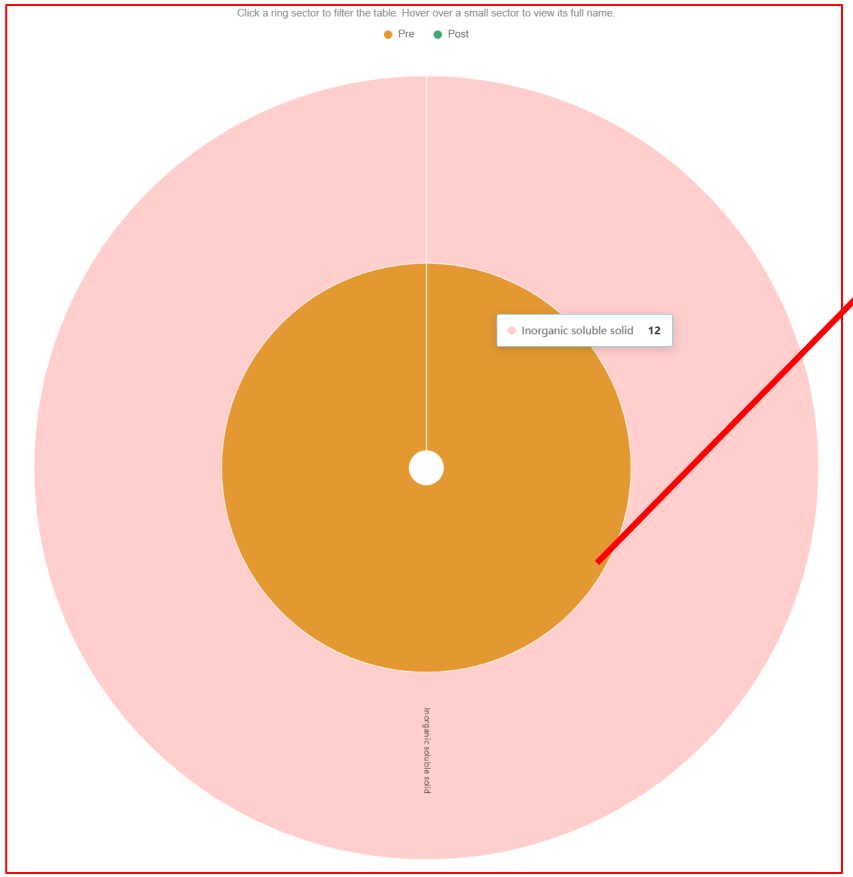
The Gene Search page within the Genomics module allows users to search by genomic region, gene ID, or gene name to retrieve information such as gene location, gene size, transcription orientation, gene structure, genomic/CDS/protein sequences, and functional annotations.

5. Phenotype module



The sunburst chart enables interactive visualization of traits. Users can click on preharvest/postharvest categories or specific traits within the chart to filter and view the corresponding phenotypic data, with the filtered results synchronously displayed in the table browser below.

5. Phenotype module



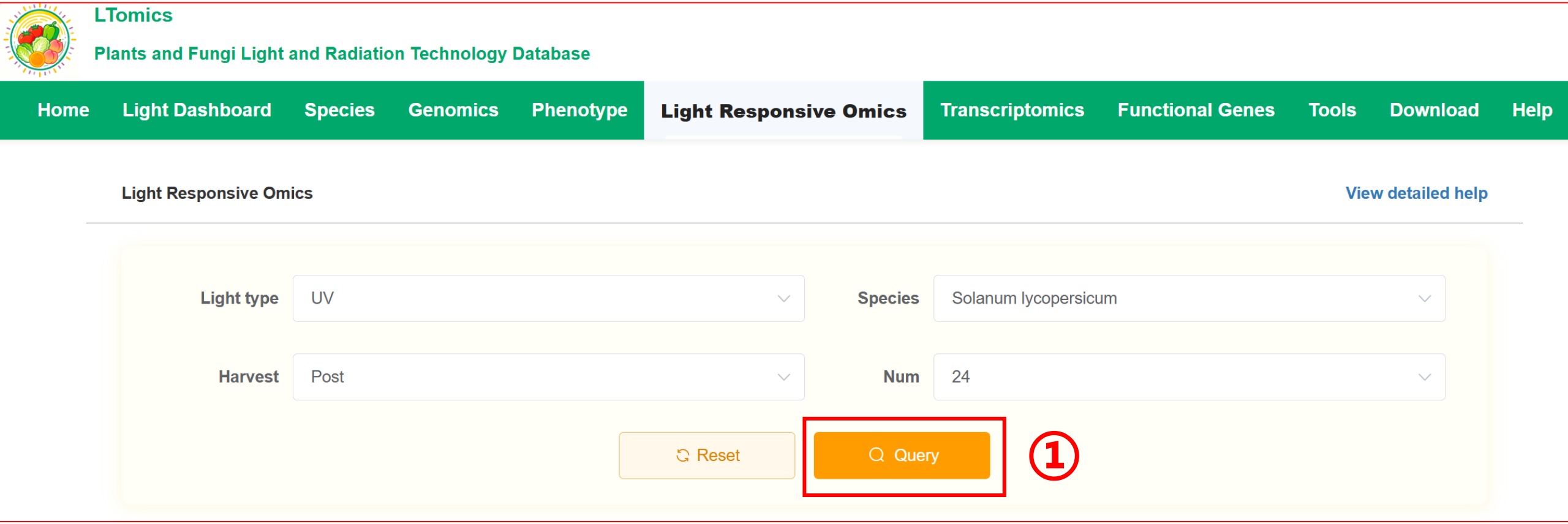
Physiological indicators

Chart selection: Pre | Inorganic soluble solid | Download

Pre/Post	Phenotype classes	Light types	Species	Dose	Storage / measurement…	Storage temper…	Unit
Pre	Inorganic soluble solid	γ-ray	Anise	5 kGy	0 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	10 kGy	0 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	15 kGy	0 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	20 kGy	0 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	5 kGy	6 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	10 kGy	6 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	15 kGy	6 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	20 kGy	6 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	5 kGy	12 months	NA	%
Pre	Inorganic soluble solid	γ-ray	Anise	10 kGy	12 months	NA	%

Clicking on "Inorganic soluble solid" under the "Pre" (Preharvest) category displays 12 phenotypic records below. The tag filters located above the table support combined filtering.

6. Light Responsive Omics module



In the Light Responsive Omics module, set Light type to UV, Species to Solanum lycopersicum (Tomato), Harvest to Post, and Num to 24, then click Query.

6. Light Responsive Omics module

Sample description ②

Sample ID	Sample	Species	Species name	Sample type	Temperature	Storage time
24-1	CK-0d	Solanum lycopersicum	Original No. 1	-	20°C	-
24-2	CK-8d	Solanum lycopersicum	Original No. 1	-	20°C	8 days
24-3	UV-C-8d	Solanum lycopersicum	Original No. 1	post-harvest 4 kJ/m2 UV-C treatm…	20°C	8 days

Transcriptome

Metabolome

Proteome

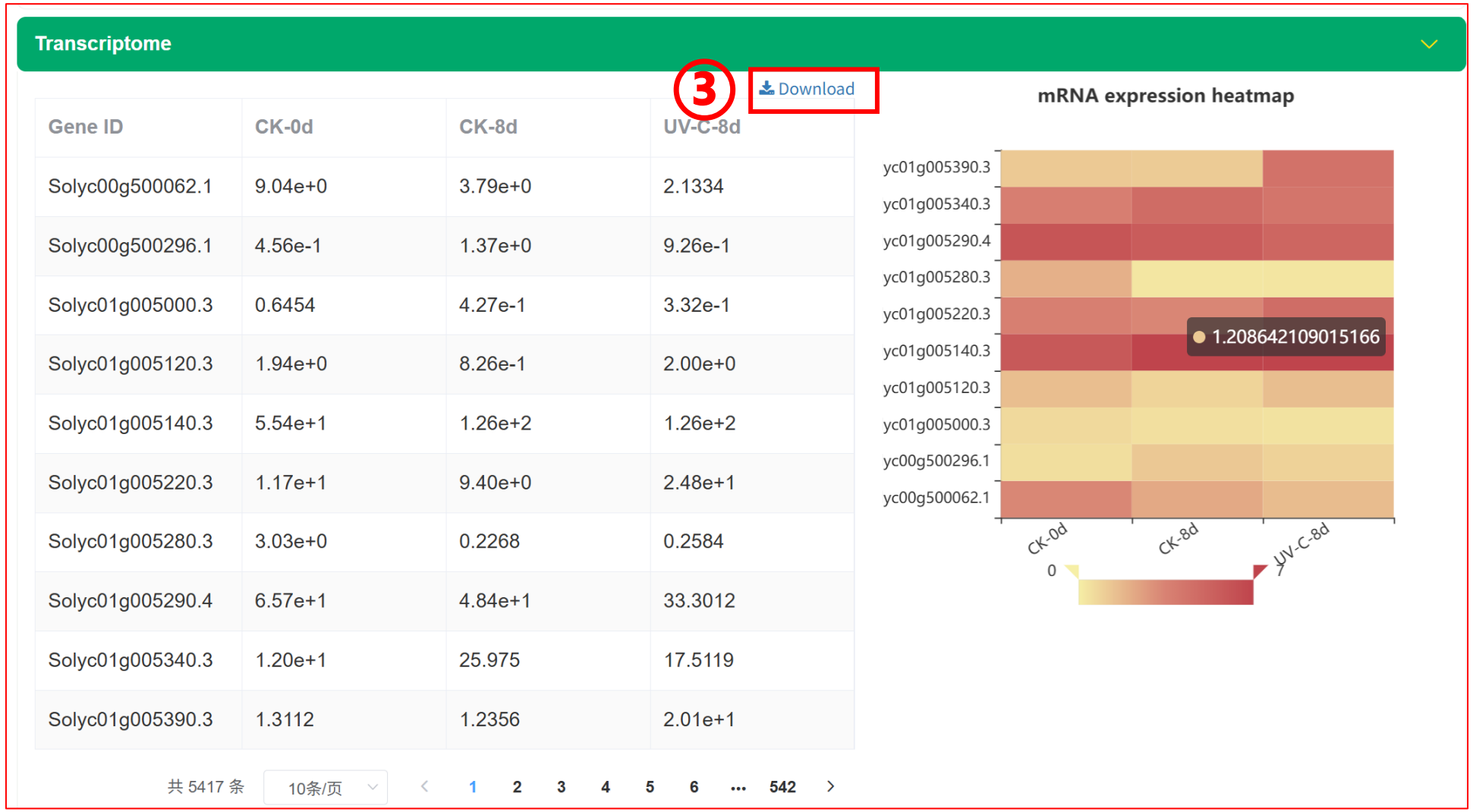
ATAC-seq

DNA methylation

RNA methylation

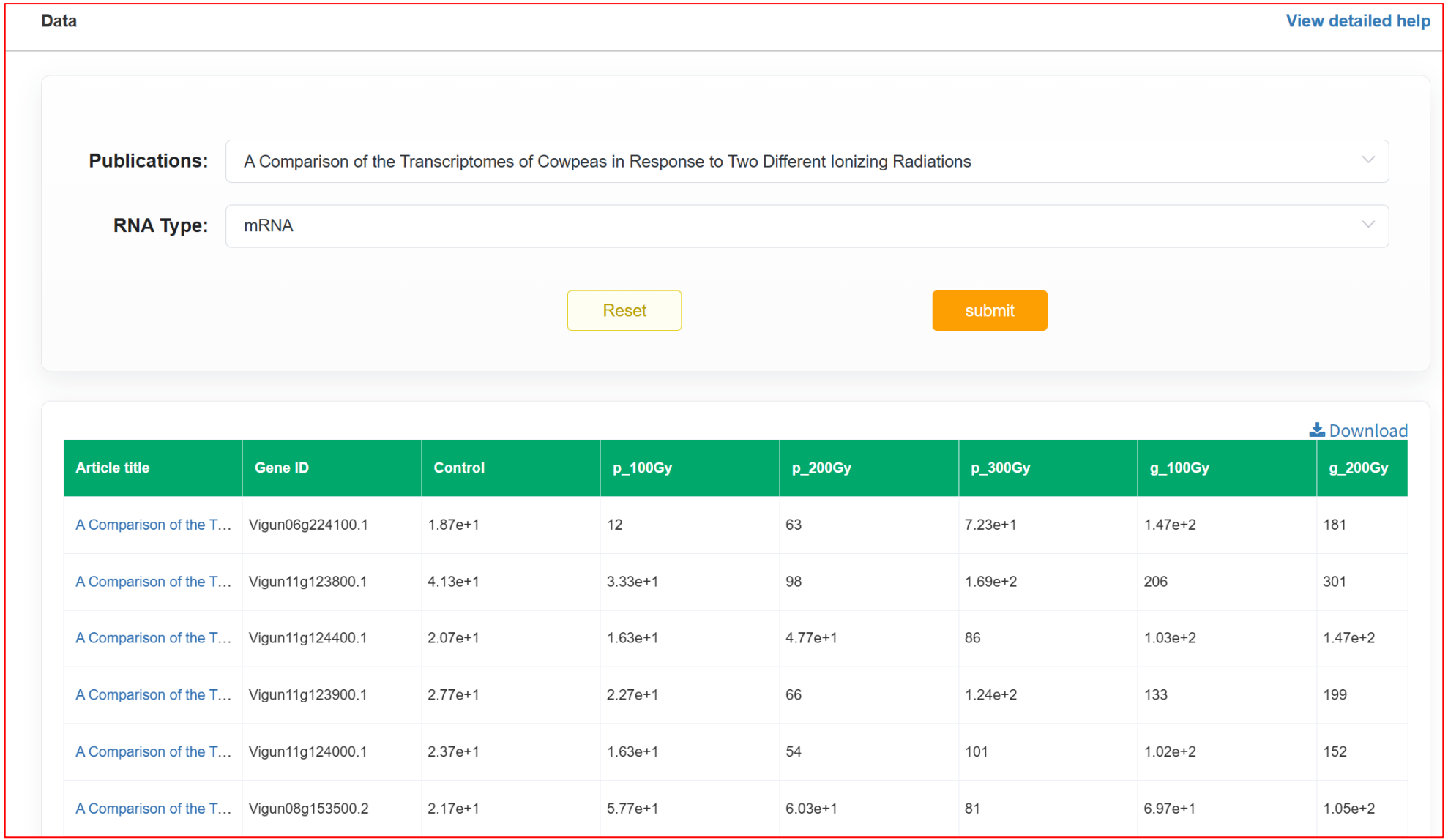
Retrieve the descriptions of different sample groups and the corresponding omics data.

6. Light Responsive Omics module



Transcriptome data as an example: data display, download, and differential analysis. Global gene expression is shown in tables and heatmaps; click Download to export all gene expression data.

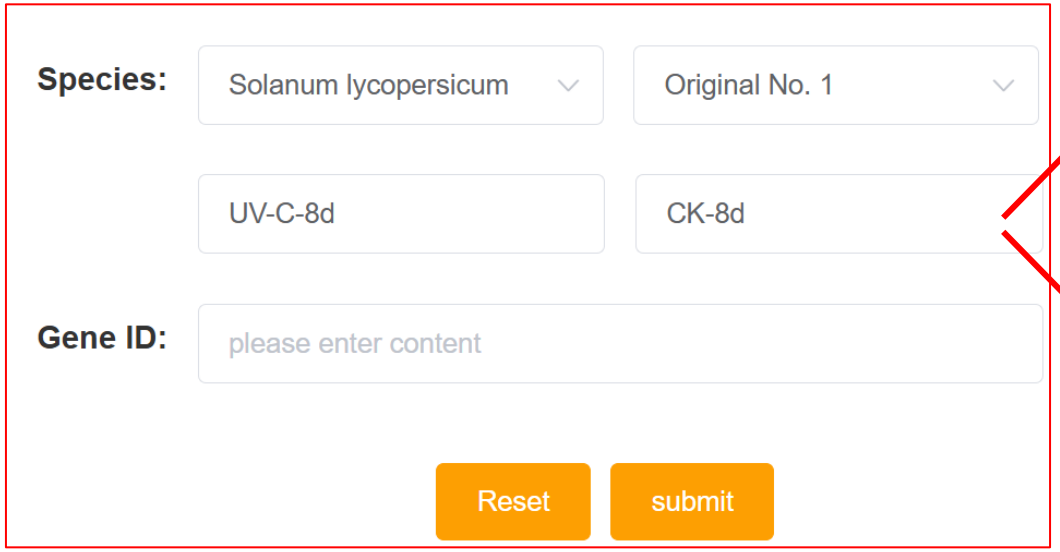
7. Transcriptomics-Data

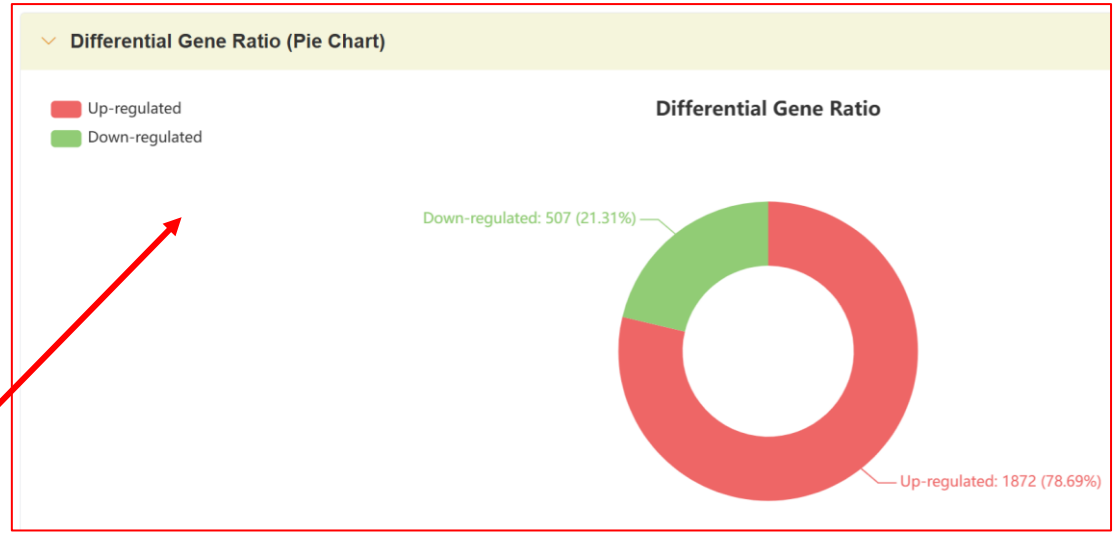


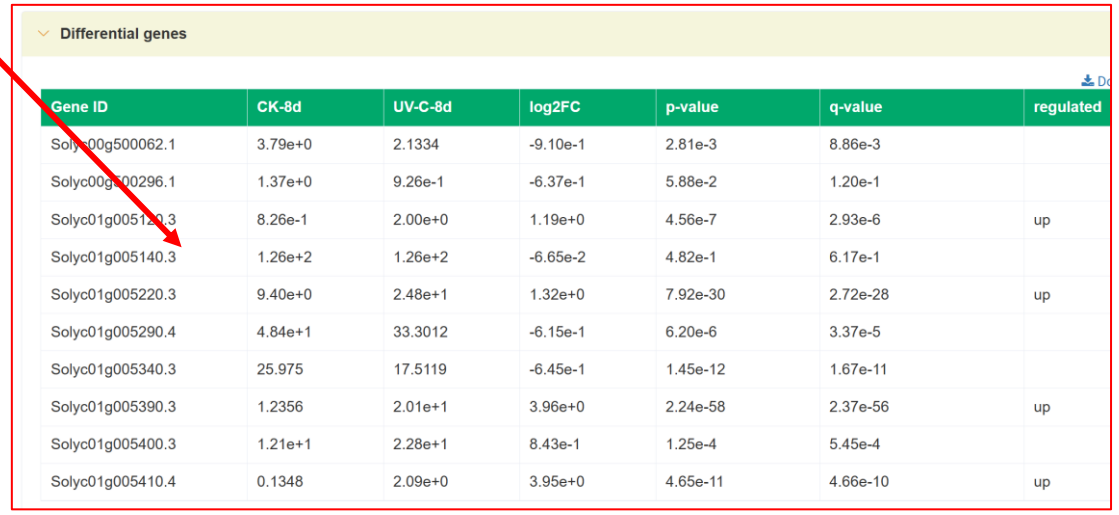
Article title	Gene ID	Control	p_100Gy	p_200Gy	p_300Gy	g_100Gy	g_200Gy
A Comparison of the T…	Vigun06g224100.1	1.87e+1	12	63	7.23e+1	1.47e+2	181
A Comparison of the T…	Vigun11g123800.1	4.13e+1	3.33e+1	98	1.69e+2	206	301
A Comparison of the T…	Vigun11g124400.1	2.07e+1	1.63e+1	4.77e+1	86	1.03e+2	1.47e+2
A Comparison of the T…	Vigun11g123900.1	2.77e+1	2.27e+1	66	1.24e+2	133	199
A Comparison of the T…	Vigun11g124000.1	2.37e+1	1.63e+1	54	101	1.02e+2	152
A Comparison of the T…	Vigun08g153500.2	2.17e+1	5.77e+1	6.03e+1	81	6.97e+1	1.05e+2

On the Data page, users can directly retrieve gene expression data for all genes by selecting specific "Publications" and "RNA Type".

7. Transcriptomics—mRNA







Gene ID	CK-8d	UV-C-8d	log2FC	p-value	q-value	regulated
Solyc00g500062.1	3.79e+0	2.1334	-9.10e-1	2.81e-3	8.86e-3	
Solyc00g500296.1	1.37e+0	9.26e-1	-6.37e-1	5.88e-2	1.20e-1	
Solyc01g005120.3	8.26e-1	2.00e+0	1.19e+0	4.56e-7	2.93e-6	up
Solyc01g005140.3	1.26e+2	1.26e+2	-6.65e-2	4.82e-1	6.17e-1	
Solyc01g005220.3	9.40e+0	2.48e+1	1.32e+0	7.92e-30	2.72e-28	up
Solyc01g005290.4	4.84e+1	33.3012	-6.15e-1	6.20e-6	3.37e-5	
Solyc01g005340.3	25.975	17.5119	-6.45e-1	1.45e-12	1.67e-11	
Solyc01g005390.3	1.2356	2.01e+1	3.96e+0	2.24e-58	2.37e-56	up
Solyc01g005400.3	1.21e+1	2.28e+1	8.43e-1	1.25e-4	5.45e-4	
Solyc01g005410.4	0.1348	2.09e+0	3.95e+0	4.65e-11	4.66e-10	up

In the mRNA subpage, users can select the target species, preharvest/postharvest treatment method, and specify the treatment and control groups to view the expression profiles of all transcripts and differentially expressed genes (DEGs) under the corresponding conditions. By entering a target gene ID, users can retrieve key information, including its expression level, fold change, and up/down-regulation direction across different samples and treatment conditions.

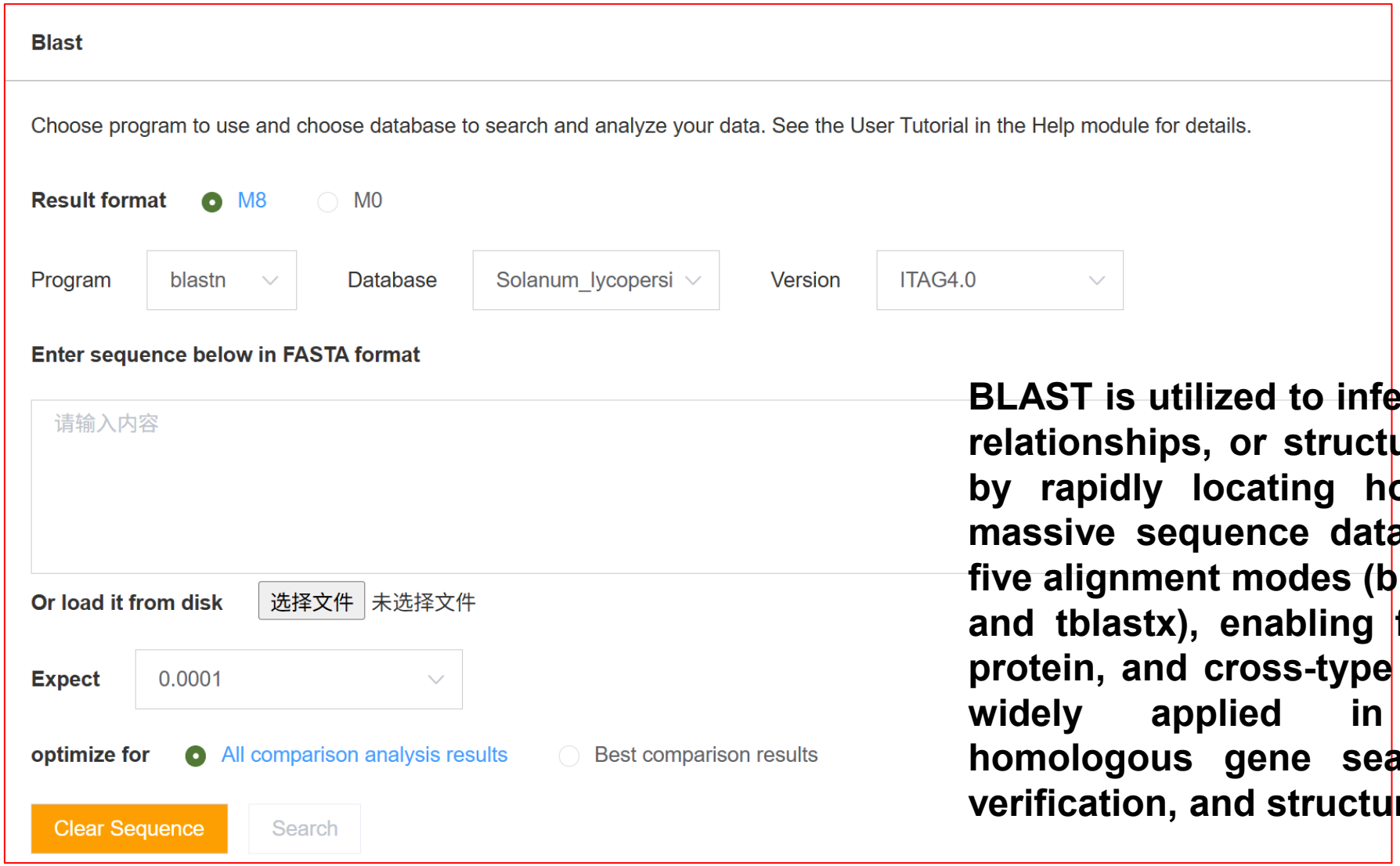
8. Functional Genes module

Result | Download | Columns

Gene ID	Gene name	Category	Species	Light type and dose	Trait category	Description	Pu
CA16g00000	CaBBX16	Light signal tra…	Capsicum ann…	Light irradiatio…	Secondary me…	CaBBX16, a B-box transcrip…	Ca
NA	BcHY5	Light signal tra…	Brassica rapa …	Purple LED lig…	Secondary me…	Purple LED irradiation durin…	Pu
NA	GSTU11	Light signal tra…	Brassica rapa …	Purple LED lig…	Secondary me…	Purple LED irradiation durin…	Pu
NA	SUR1	Light signal tra…	Brassica rapa …	Purple LED lig…	Secondary me…	Purple LED irradiation durin…	Pu
NA	JsHY5	Secondary me…	Juglans sigillat…	High light irradi…	Secondary me…	This study reveals that light i…	Li
NA	JsMYB141	Secondary me…	Juglans sigillat…	High light irradi…	Secondary me…	This study reveals that light i…	Li
NA	JsFLS4	Secondary me…	Juglans sigillat…	High light irradi…	Secondary me…	This study reveals that light i…	Li
NA	PbFAR1	Transcription f…	Phoebe bournei	Light irradiatio…	Light response…	This study conducted a gen…	G
NA	PbFHY3	Transcription f…	Phoebe bournei	Light irradiatio…	Light response…	This study conducted a gen…	G
NA	PbFRS	Transcription f…	Phoebe bournei	Light irradiatio…	Light response…	This study conducted a gen…	G

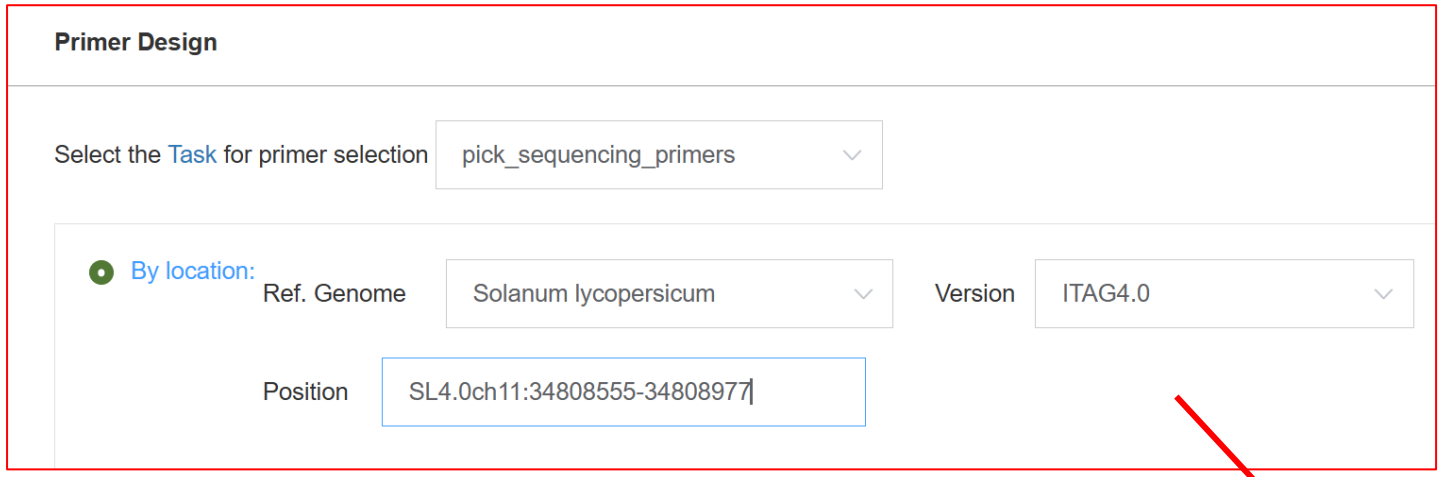
The Functional Gene module includes 140 light- and irradiation-responsive genes, which are individually graded based on Table 3. Specifically, 119 genes are validated by genetic functional experiments (V3–V4 levels), while 21 genes are supported only by expression profiling, association analysis, or bioinformatics predictions (V1–V2 levels, considered candidate genes).

9. Tools module—Blast



BLAST is utilized to infer the function, evolutionary relationships, or structure of unknown sequences by rapidly locating homologous regions within massive sequence databases. This tool supports five alignment modes (blastn, blastp, blastx, tblastn, and tblastx), enabling flexible DNA-DNA, protein-protein, and cross-type sequence alignments. It is widely applied in functional prediction, homologous gene searching, primer specificity verification, and structural variation analysis.

9. Tools module—Primer Design

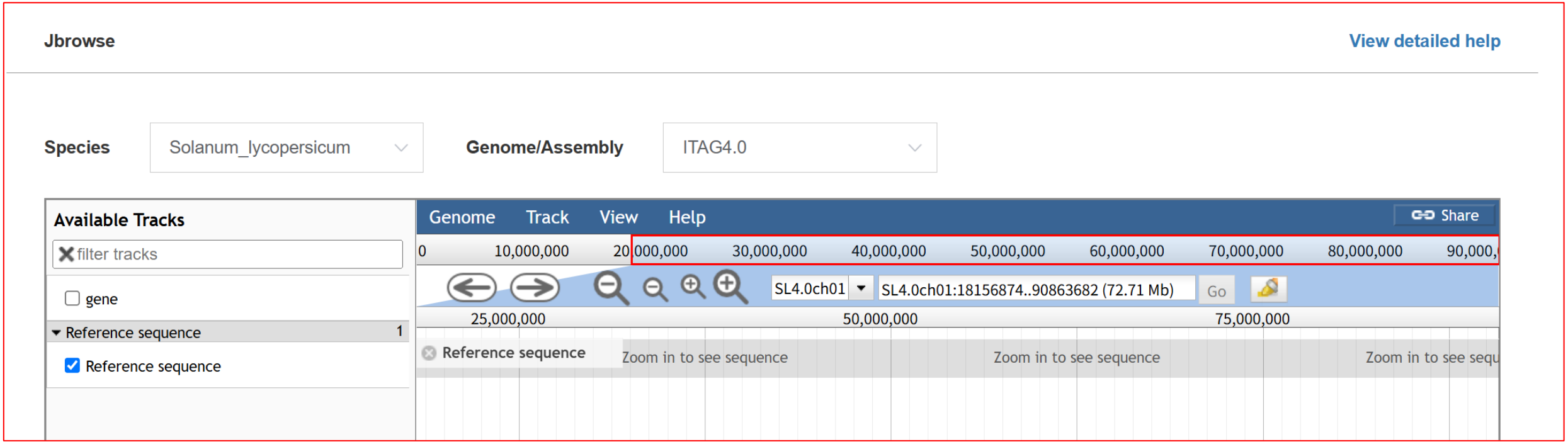


	start	len	tm	gc%	any	3'	seq
1 LEFT_PRIMER	3	20	59.96	55.00	4.00	3.00	GGTCACGCCCCCTTCTTAAA
2 LEFT_PRIMER	308	20	58.88	50.00	2.00	0.00	TTTTTGCCTTGAGTGGTGGG

	start	len	tm	gc%	any	3'	seq
1 RIGHT_PRIMER	189	23	58.86	39.13	5.00	1.00	TGCATTGCTGCATATCATCTTCT
2 RIGHT_PRIMER	483	20	57.60	50.00	6.00	2.00	GTAGTGCGCATCGTGATCTA

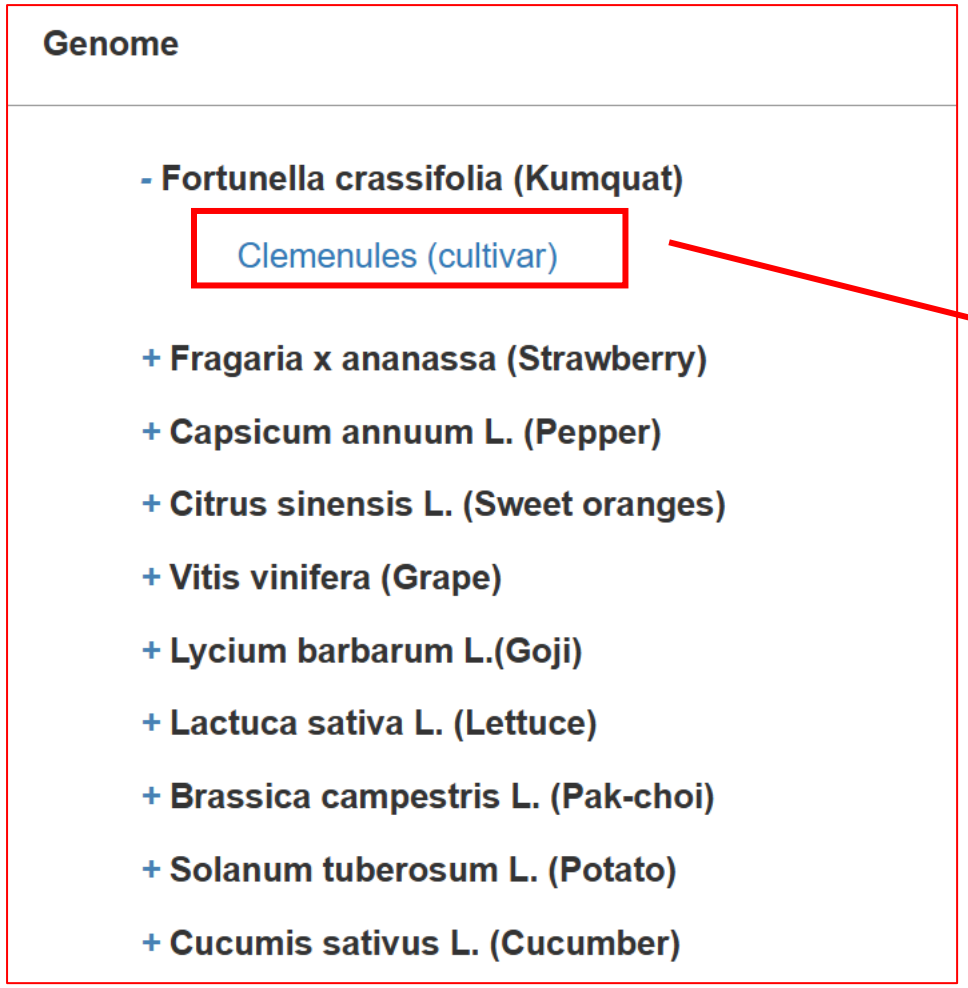
The Primer Design tool offers two modes: (1) Genome position-based design: Users select a species and specify the chromosome/scaffold and coordinates, then adjust parameters (amplicon size, Tm, GC content, primer length) to get candidate primer pairs with sequences, product lengths, Tm, GC content, and secondary structure predictions. (2) Sequence-based design: Users input sequences (FASTA/plain text, single/batch) and set custom parameters. The tool performs specificity screening against the background genome or internal algorithm validation, providing recommended primer pairs and evaluation metrics suitable for qRT-PCR, genotyping, and amplicon sequencing.

9. Tools module—Jbrowse



After selecting a target species, users can intuitively view gene annotations, expression tracks, and sequence features in a linear genome view, utilizing smooth navigation, zoom controls, and mouse hover/click functions.

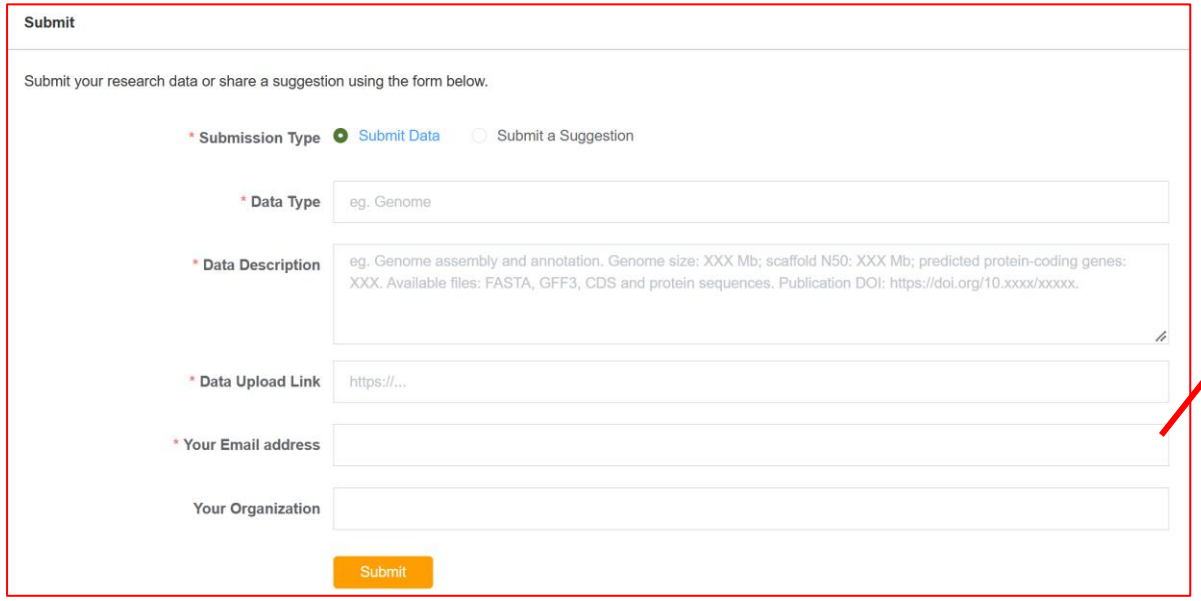
10. Download module



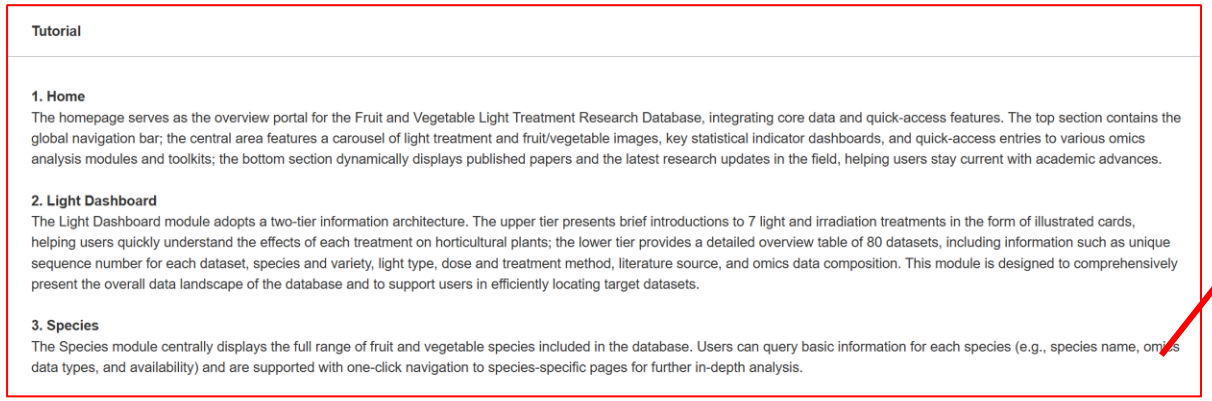


The Download module allows users to retrieve data from 41 species via dropdown selection or one-click download. All files are provided in standard formats (FASTA, GFF3, expression matrices, and differentially expressed genes) to facilitate seamless downstream analysis.

11. Help module



This gateway serves as an interactive portal between the platform and its users, primarily for data submission and error feedback. Through this portal, users can report issues or bugs encountered during use to the maintenance team, or submit new datasets and literature to help the database continuously update and expand.



This is a key gateway for new users. It provides step-by-step operational guides, instructing users on how to utilize various platform functions, such as browsing data, performing BLAST alignments, and designing primers. Serving as the platform's "user manual," it is designed to help users quickly familiarize themselves with the environment and complete their analysis workflows.



"Cite" provides a standardized citation format, making it convenient for users to accurately cite this database platform in their academic publications.